Problem Set 10.

1). $T \in L(V)$

To prove: if T has upper $\triangle$ matrix wrt basis $(V) : \{v_1, \ldots, v_n\}$

T is upper $\triangle$ wrt $\{e_1, \ldots, e_n\}$

$e_1 := v_1$ for $j \in [1, n]$.

$\therefore e_j := \underbrace{(v_j - \langle v_j, e_1 \rangle e_1 - \cdots}_{c_j}$

Proof:

T is upper $\triangle$ wrt $\{e_1, \ldots, e_n\}$
$\equiv T(e_j) \in \mathrm{span}(e_1, \ldots, e_j) \quad j = 1$

Now, $e_j = \underbrace{v_j - \sum_{k=1} (v_j, e_k) e_k}_{c_j}$

$\therefore e_j \in \mathrm{span}(v_1, \ldots, v_j\}$

$\therefore e_j = \sum_{i=1}^{j} \alpha_i v_i$

$\therefore T(e_j) = \sum T(\alpha_i v_i) = \sum \alpha_i T(v_i)$.

$T(v_i) \in \text{span}(v_1, \ldots v_i) \subseteq \text{span}(v_1, \ldots v_j)$

$\therefore$ its upper $\triangleleft$ wrt basis (V)

~~$\therefore T(v_i) \subseteq \text{span}(v_1, \ldots$~~

$\therefore T(e_j) \in \text{span}(v_1, \ldots v_j)$

Now,

$\text{span}(v_1, \ldots v_j) = \text{span}(e_1, \ldots e_j)$

$\therefore T(e_j) \in \text{span}(e_1, \ldots e_j)$.

$\therefore T$ has an upper $\triangleleft$ matrix in the orthonormal basis $\{e_1, \ldots e_n\}$

2). Given: V is a complex inner product space.

To prove:

$$\langle u, v \rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i}{4}$$

$\forall\, u, v \in V$.

Proof: $\|x\|^2 = \langle x, x \rangle$

$\therefore \|u+v\|^2 = \langle u+v, u+v \rangle$

$$= \boxed{\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle} \quad (1)$$

$\|u-v\|^2 = \langle u-v, u-v \rangle$

$$= \boxed{\langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle} \quad (2)$$

$\therefore$ from (1) & (2),

$\|u+v\|^2 - \|u-v\|^2$

$+ 2\langle u, v\rangle + 2\langle v, u\rangle$

$\langle v, u\rangle = \overline{\langle u, v\rangle}$

$2\langle u, v\rangle + 2\overline{\langle u, v\rangle}$

$= \boxed{4\, \mathrm{Re}(\langle u, v\rangle)}$

$\|u + iv\|^2 = \langle u + iv, u + iv\rangle$

$= \langle u, u\rangle + i\langle u, v\rangle - i\langle v, u\rangle + \langle v, v\rangle$ — (3)

$\|u - iv\|^2 = \langle u - iv, u - iv\rangle$

$= \langle u, v\rangle - i\langle u, v\rangle + i\langle v, u\rangle + \langle v, v\rangle$

— (4)

Substracting (4) from (3),

$\|u + iv\|^2 - \|u - iv\|^2 = 2i\langle u, v\rangle - 2i\langle v, u\rangle$

$= 2i\,(\langle u, v\rangle - \overline{\langle u, v\rangle})$

$= \cancel{2i}\ 4\, \mathrm{Im}(\langle u, v\rangle)$

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$\therefore \cancel{\frac{1}{4}\left(4(Re(\langle u,v\rangle)\right)} + i\,4\,Img \cancel{3}$

$\therefore \cancel{\frac{1}{4}}$

$$\therefore \frac{1}{4}\left(4\,Re\langle u,v\rangle + i\left(4\,Im(\langle u,v\rangle)\right)\right)$$

$$= \langle u,v\rangle$$

$$\therefore \boxed{\langle u,v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2}{4}}$$

Hence proved.

$\langle v,v\rangle$

$v\rangle$

$v,w\rangle$

$v\rangle)$

3). On $P_2(\mathbb{R})$,

$$\langle u,v \rangle = \int_0^1 p(x)\, q(x)\, dx.$$

To find: orthonormal basis of $P_2(\mathbb{R})$.

Sol^n.

$V = \{1, x, x^2\}$

$v_1 = 1$

$$\therefore \|1\|^2 = \int_0^1 1\,dx = [x]_0^1 = 1-0 = 1$$

$$\therefore \boxed{e_1 = 1}$$

$v_2 = x - \langle x, e_1 \rangle e_1$

$$\langle x, 1 \rangle = \int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1$$

$$\therefore \boxed{\frac{1}{2}}$$

$$\therefore v_2 = x - \frac{1}{2}$$

$$\therefore \|v_2\|^2 = \int_0^1 \left(x - \frac{1}{2}\right)^2 dx$$

$$= \int_0^1 \left(x^2 - x + \frac{1}{4}\right) dx$$

(※):

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{4}\right]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{4}$$

$$= \boxed{\frac{1}{12}}$$

$$\therefore e_2 = \frac{x - \frac{1}{2}}{\sqrt{1/12}} = \sqrt{12}\left(x - \frac{1}{2}\right)$$

$$= 2\sqrt{3}\left(x - \frac{1}{2}\right)$$

$$= \boxed{\sqrt{3}(2x-1)}$$

for $v_3 = x^3$

$$v_3 = x^2 - \langle x^2, e_1\rangle e_1 - \langle x^2, e_2\rangle e_2$$

$$\langle x^2, 1\rangle = \int_0^1 x^2 dx = \frac{1}{3}$$

$$\langle x^2, e_2\rangle = \sqrt{3}\int_0^1 x^2(2x-1)dx$$

$$= \sqrt{3}\left[\frac{2x^4}{4} - \frac{x^3}{3}\right]_0^1$$

$$= \sqrt{3}\left[\frac{1}{2} - \frac{1}{3}\right]$$

$$\boxed{= \frac{\sqrt{3}}{6}} = \boxed{\frac{\sqrt{12}}{12}}$$

$$\therefore v_3 = x^2 - \frac{1}{3} - \frac{\cancel{\sqrt{12}}}{\cancel{12}}\left(\cancel{\sqrt{12}}\left(x - \frac{1}{2}\right)\right)$$

$$= x^2 - \frac{1}{3} - x + \frac{1}{2}$$

$$= \boxed{x^2 - x + \frac{1}{6}}$$

$\frac{x^3}{3} + x^2$

$\therefore \|v_3\|^2 = \int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx$

$= \int_0^1 \left(x^4 - x^3 + \frac{x^2}{6} - x^3 + x^2 - \frac{x}{6} + \frac{x^2}{6} - \frac{x}{6} + \frac{1}{36}\right) dx$

$= \int_0^1 \left(x^4 - 2x^3 - \frac{x}{3} + \frac{1}{36} + \frac{4x^2}{3}\right) dx$

$= \frac{1}{180}$

$\boxed{\therefore e_3 = \sqrt{180}\left(x^2 - x + \frac{1}{6}\right)}$

$\therefore$ orthonormal basis,

$\boxed{\left\{ 1, \sqrt{12}\left(x - \frac{1}{2}\right), \sqrt{180}\left(x^2 - x + \frac{1}{6}\right) \right\}}$